

Oluwatimilehin Sapara

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EDUCATION

Texas Tech University

Lubbock, TX

Bachelor of Arts in Computer Science, Minor in Math and Actuarial Science

Expected Graduation: December 2027

- **GPA: 4.0**
- **Relevant Coursework:** Programming Principles, Discrete Math, Data Structures and Algorithms, Object-Oriented Programming, Concepts of Programming Languages, Data Science, Theory of Automata, Assembly Language Programming, Technical Writing, Linear Algebra, Modern Digital System Design.

TECHNICAL SKILLS

Languages: JavaScript, Python, C/C++, Java, SQL (Postgres), HTML/CSS, MATLAB

Frameworks: React, Next, Node.js, Three.js, Flask, PyTorch, Tailwind, FastAPI

Developer Tools: Vercel, Git, AWS, Django, Vite, Azure Lab Services, Google Cloud Platform, IntelliJ

Libraries: Pandas, NumPy, Matplotlib, scikit-learn, TensorFlow, XGBoost, SciPy

PROJECTS

Cooke Language Compiler | *C, GCC, C17*

March 2026 – Present

- Designed and implemented a lexical analyzer for a custom programming language, handling tokenization of 25+ token types, operator disambiguation, and identifier parsing.
- Building a recursive-descent syntax parser supporting 8+ custom operators and word-based boolean logic.

Jobify | *React, JavaScript, HTML, TailwindCSS*

July 2025 – August 2025

- Built a responsive job application platform with full CRUD functionality in React.
- Integrated a custom RESTful API using json-server for data persistence, client-side routing, and form validation.
- Reduced application errors by 25% and improved load times by 30% for 15+ active test users.

ML-Driven Concrete Strength Predictor | *Python, Pandas, Matplotlib, scikit-learn*

June 2025

- Designed and trained ML models (Linear Regression, Random Forest, Ridge) to predict compressive strength across 1000+ samples.
- Achieved 92% accuracy through cross-validation, hyperparameter tuning, and feature engineering on structural material data.
- Developed Matplotlib dashboards to visualize model performance, feature importance, and prediction error distributions.

EXPERIENCE

Operations Analytics Co-op | *Delta Air Lines*

May 2026 – August 2026

- Built a full-stack KPI visualization tool (Python, Streamlit, Plotly, NetworkX) supporting 300+ KPIs, replacing static spreadsheet reporting with interactive drill-down views for leadership.
- Engineered a goal-cascading algorithm that propagates targets across hierarchies of 20,000+ nodes in under 30 seconds, with automated validation ensuring rollup accuracy at every level.
- Designed a configurable hierarchy system (Tkinter GUI + JSON schema) letting non-technical stakeholders redefine reporting structures without code, backed by a caching layer to reduce warehouse load.

Undergraduate Research Assistant | *Texas Tech University*

April 2025 – Present

- Developed an XGBoost-based multi-step time-series model for pressure prediction across atmospheric field data, reducing inference runtime from 3 hours → 20 minutes.
- Processed and analyzed 50+ datasets of wind tunnel and large-eddy simulation (LES) data using Python and MATLAB to validate computational models of tornado-like vortices (TLVs).
- Achieved 1.95 RMSE on held-out validation data using cubic spline interpolation and hyperparameter tuning.

LEADERSHIP & VOLUNTEERING

ColorStack TTU - Outreach Committee Member

August 2024 – Present

- Engaged over 100 students through coding workshops, mentorship programs, and networking initiatives to increase participation in computer science.

National Society of Black Engineers (NSBE) - Active Member

January 2024 – Present

- Volunteered at campus events and STEM outreach programs encouraging K-12 students to pursue careers in STEM fields.